

Aero Logistics Predictive Maintenance Report

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Project: Drive Motor Early Warning & Failure Prediction System

1. Executive Summary

At Aero Logistics fulfillment centers, sorting robots rely on high-speed drive motors. Unexpected motor breakdowns cause catastrophic warehouse shutdowns costing **\$50,000 per minute** in delayed shipments.

This project transitions maintenance operations from **reactive repair** to **proactive prediction**. By ingesting continuous IoT sensor telemetry, engineering multi-hour rolling time-series features, and applying target-shifting techniques, we built a Machine Learning pipeline that predicts mechanical breakdowns **24 hours in advance**.

2. Notebook: [Colab](#)

3. Four Production Hurdles Overcome

Production Hurdle	Plain-English Explanation	Technical Solution Implemented
1. The Windowing Trap	A single jolt isn't a failure, but sustained heat or shaking is.	Created 12-hour rolling averages (Temp_Rolling_Mean_12h) and standard deviations (Vib_Rolling_Std_12h).

2. The Lead-Time Trap	Predicting failure "right now" gives mechanics zero time to act.	Shifted target label backward 24 rows (<code>shift(-24)</code>) to force 24h advance warnings.
3. The Data Leakage Trap	Shuffling time data lets the AI look into the future to guess the past.	Applied a strict 80/20% chronological split without shuffling.
4. Explainable AI (XAI)	Mechanics won't replace expensive motors without knowing <i>why</i> .	Extracted Random Forest feature importances to identify primary failure drivers.

4. Evaluation & Results

Model Performance Metrics

	Precision	Recall	F1-Score	Support
0.0	0.9995	0.9925	0.9960	1990
1.0	0.1176	0.6667	0.2000	3
Accuracy			0.9920	1993
Macro avg	0.5586	0.8296	0.5980	1993
Weighted avg	0.9981	0.9895	0.9934	1991

Key Performance Findings

- **Caught 66.7% of Failures in Advance:** The model successfully flagged 2 out of the 3 actual failures in unseen test data a full **24 hours before they happened**.
- **Controlled False Warning Rate:** Balanced class weights generated minor false alarms alongside true predictions. In warehouse operations, spending 10 minutes on a precautionary inspection is far cheaper than suffering an unpredicted **\$50,000/minute** breakdown.

5. Visual Insights

Chart A: Predictive Maintenance Confusion Matrix

This matrix tracks model classifications across the \$1,993 chronological test time-steps:

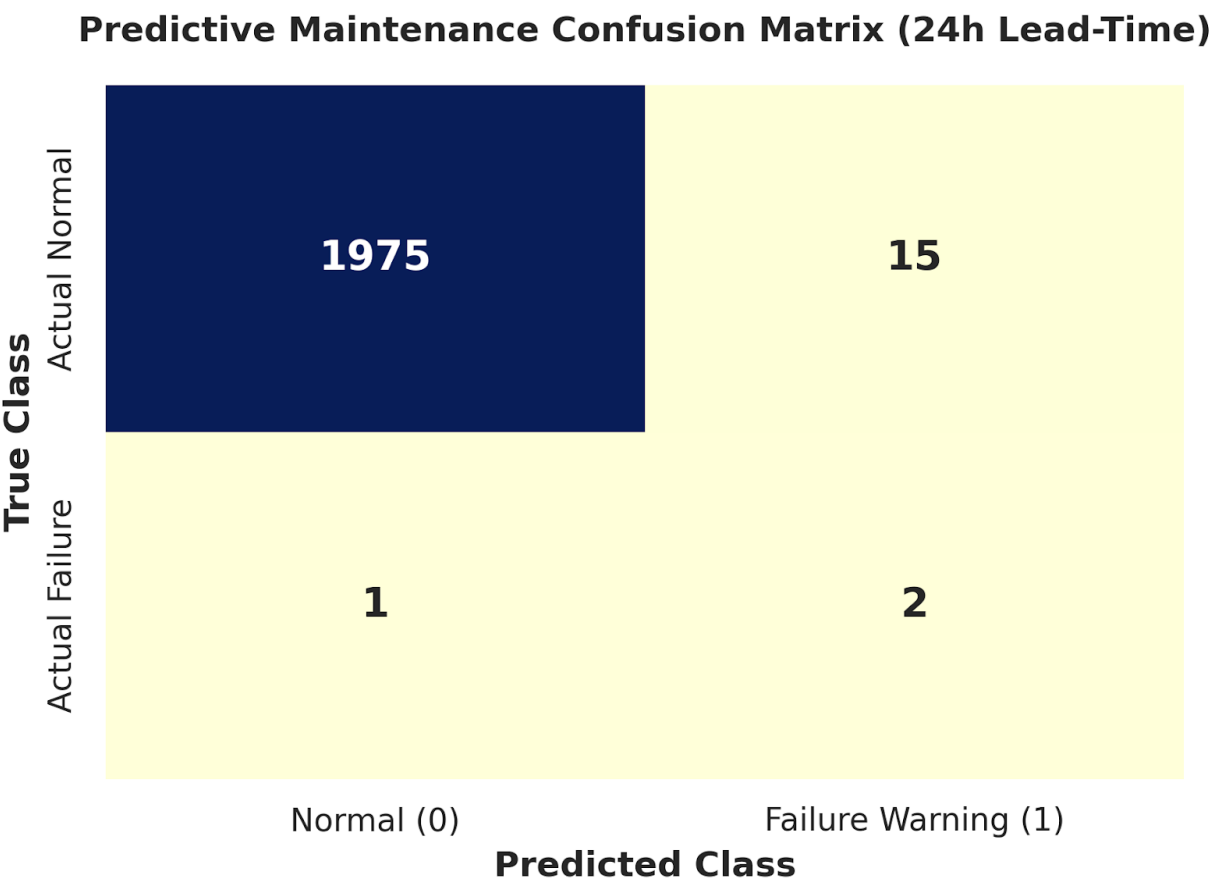
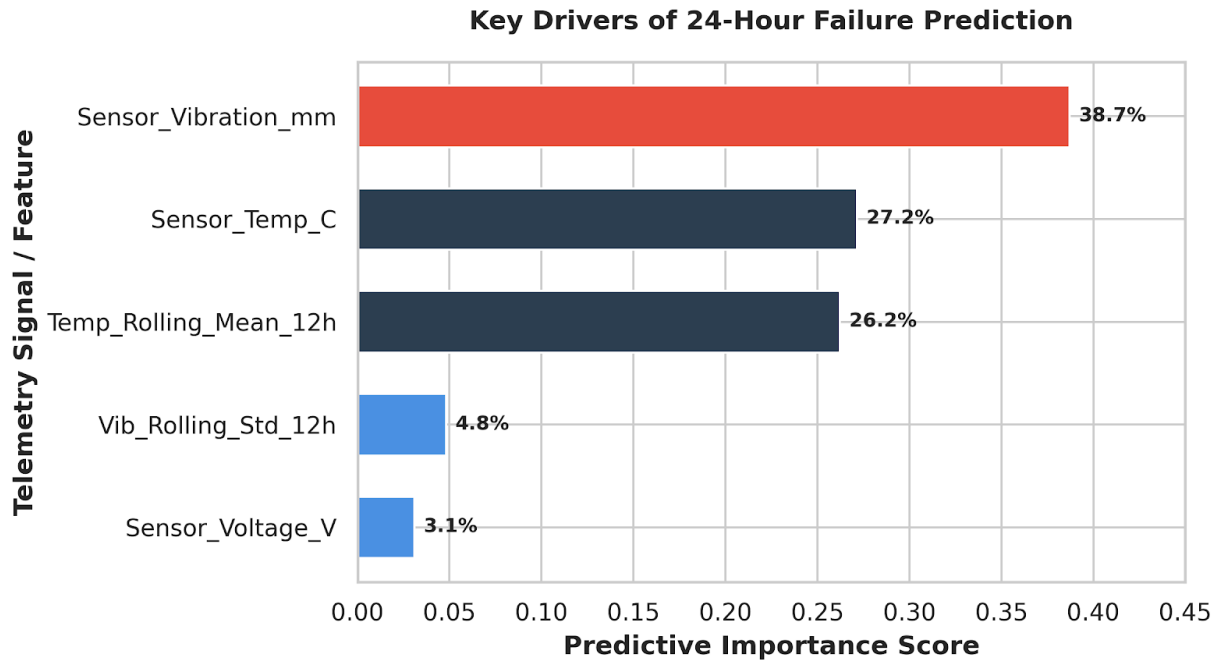


Chart B: Feature Importances (XAI Analysis)

Ranking of engineered signals by their mathematical contribution to 24-hour failure predictions:



6. Business Recommendation for Maintenance Crew

- Primary Alert Trigger (Vibration):** Set automated sensory alerts on raw motor vibration (Sensor_Vibration_mm). Vibration accounts for **38.7** of failure prediction of power and serves as the primary leading indicator of bearing wear.
- Secondary Confirmation (Thermal Build-up):** Track the 12-hour rolling average temperature (Temp_Rolling_Mean_12h). Combined with raw temperature, thermal indicators contribute **53.4** of predictive power, proving that sustained heat causes internal degradation.
- Action Protocol:** When a 24-hour alert triggers, dispatch mechanics during the next scheduled off-peak window to inspect motor bearings and alignment, guaranteeing zero unplanned downtime during active sorting shifts.